

Thomas Zhang

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

- Percussionist at Orchestra@UWaterloo

Waterloo, ON

Sep. 2025 - Apr. 2030 (Expected)

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript/TypeScript, Swift, Java, C#, SQL, HTML/CSS

AI/ML: PyTorch, OpenCV, NumPy, OR-Tools, Cloudflare Workers AI, Vectorize, Pinecone

Cloud & Infrastructure: AWS (EC2, S3, ALB, EBS, EFS, Fargate, CloudWatch), Cloudflare (Workers, Durable Objects, D1, R2, Browser Rendering), Terraform, Kubernetes, Docker, GitHub Actions, GitLab CI/CD, Datadog

Backend & Data: FastAPI, Flask, Node.js, React.js/Next.js, React Native/Expo, Apache Kafka, Redis, Celery, TimescaleDB, PostgreSQL, Supabase, SQLite

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Sep. 2026

Stealth Startup

Toronto, ON

- Built a sandboxed autonomous agent harness that lets engineers and non-engineers ship by filing a ticket, returning green PRs gated by unit, UI, integration, and e2e testing on a **Docker Compose** replica of the company stack
- Autonomously resolved **40+** tickets with **90%+** success while cutting cost **6x/issue** (**<\$3**), saving **10+** hrs/week
- Recycled a **21**-container **AWS EC2** stack nightly, self-monitored via systemd, **CloudWatch**, and **Datadog**
- Added a self-healing loop on a durable **SQLite** queue that repairs failed CI builds and recovers dropped requests
- Authored **Terraform** to register the service on the company's shared **AWS** via **ECS** and a webhook relay
- Solely owned the service end to end across **5** codebases, scoping requirements with engineers and stakeholders

Software Engineer - Core Member

Jan. 2026 – May. 2026

Watonomous Design Team

Waterloo, ON

- Reduced obstacle costmap input from 307K to ~15K pts/frame via **20×** **point cloud** downsampling
- Added line-of-sight smoothing to **A*** planner on 60×60 grids, cutting redundant waypoints
- Integrated **YOLOv8** via **ONNX Runtime** on simulated depth camera feeds for obstacle detection

Software Engineer Intern

Jun. 2025 – Aug. 2025

Triple J Canada Consulting Inc.

Mississauga, ON

- Built a production full-stack tax platform for **2,000+** clients and **5,000+** returns, replacing paper intake and in-person office visits with web-based data collection
- Delivered **Flask** backend that reduced staff case prep work by **>30%** by shifting data entry to clients
- Streamlined onboarding with client email verification and admin RBAC, eliminating **20+ hrs/week** of work
- Ran live **SQLite** database migrations using Alembic, with **zero loss or downtime**

PROJECTS

Scale | *TypeScript, Cloudflare, Swift, Expo, React Native, Python, OR-Tools*

- Built Scale, an app that scans a real room and objects with an iPhone, then uses an AI agent to search, check fit, and propose furniture, with **OR-Tools** to compute placement so nothing overlaps, live in a VR headset
- Deployed backend with **Cloudflare Workers** and **Durable Object**, reaching a **Docker** solver via secure tunnels
- Embedded 3D models as vectors in **Vectorize** via **Browser Rendering**, with glb files in **R2** and metadata in **D1**
- Integrated native **Swift** scanning with ARKit in **Expo** app, reconstructing 3D meshes at scale with **LiDAR**

TradeStream | *C++, Python, FastAPI, Next.js, Apache Kafka, Redis, TimescaleDB, Docker*

- Architected a real-time market data platform processing **1M+** **daily ticks** across 100 symbols, streaming live VWAP, volatility, and volume-spike analytics to a **Next.js** dashboard via **WebSockets**
- Ran a **C++** microservice computing rolling analytics over a 20-tick sliding window via librdkafka
- Decoupled tick ingestion, analytics, and WebSocket broadcast across three independent **Kafka** consumer groups
- Routed live ticks to **Redis** for recent access and **TimescaleDB** hypertables for historical retention